

PM Controller

Phase 2 — Action plan · the task board, columns, drag, role-gated transitions

Owner Apex Plumbing and Mechanical Services SC **Milestone** A Lead cannot move a task to Accomplished

Target iPad first · iOS 18 **Issued** 21 September 2026

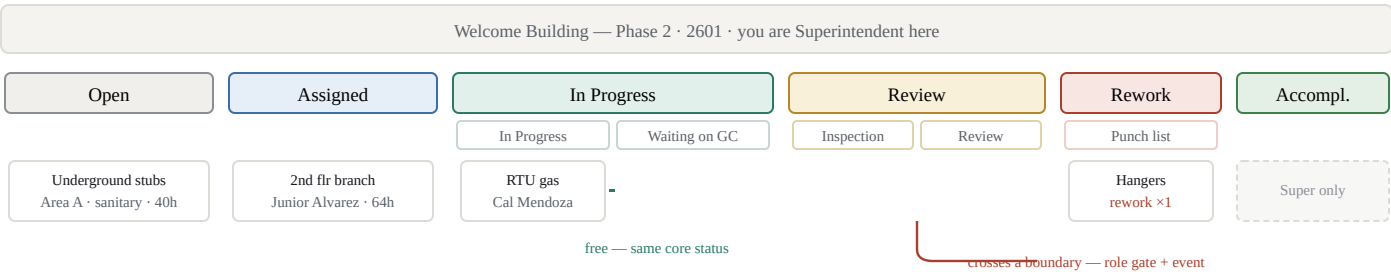
1. What Phase 2 is, and what it is not

SPEC §12 gives Phase 2 one line: *task board, columns, drag, role-gated transitions*, done when a *Lead cannot move a task to Accomplished*. That milestone is already true in the database — assertion 25 of the Phase 1 suite proves it. **Phase 2's job is to make it true through the app**, and to make the board something a Super can actually work from on a tablet.

IN SCOPE	EXPLICITLY NOT, AND WHERE IT LIVES
• Board of columns, tasks grouped beneath them • Drag a card between columns • Role gates on every transition, hidden in the UI and enforced in the database • Task detail — open a card, read it, act on it • Create a task against project, area, system • Assign to a Foreman, then to a Lead • Every move writes to <code>task_events</code>	• Managing custom columns — Phase 6a. Existing ones render; nothing creates or edits them yet. • Time capture and crew selection — Phase 3 • Photos, quantity, close-out — Phase 4 • Rework reasons and push — Phase 5 • Blocked-days clock on blocked columns — Phase 6b • The progress page — Phase 7

The rule that shapes the whole screen. Custom columns sit on top of six locked core statuses. Dragging inside one core status is free. Crossing a boundary fires the role gate and writes an event. The board must make that distinction obvious without explaining it — a card that cannot cross should not look draggable across.

2. The board, drawn



A Lead sees this same board. The Accomplished column refuses his card, the cards that are not his do not lift, and the database refuses him regardless — the UI only saves him the round trip.

3. Build order

#	STEP	DONE WHEN
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1	Board data layer — load columns and tasks for a project, grouped	The seeded Welcome Building renders its ten columns with its three tasks in the right places
2	Board layout, tablet first	Ten columns scroll horizontally on an iPad without the cards becoming unreadable
3	Task detail sheet	Tapping a card shows title, area, system, estimate, who it is assigned to, and its event history
4	Transition engine — one place that decides what a move means	Moving to a column derives the new core status from that column, never the other way round
5	Role gates in the UI	A Lead cannot begin a drag toward Accomplished; a Foreman sees no sign-off action
6	Drag and drop	A card moves between columns and the change survives a reload
7	Create and assign	A Super creates a task, assigns a Foreman, the Foreman assigns a Lead
8	Optimistic move with rollback	A refused move snaps back and says why, rather than lying about having worked
9	Tests extended	New assertions for column-to-status derivation and in-status moves; screenshots cover the board as Super and as Lead

4. Decisions to make while building

QUESTION	LEANING, AND WHY
Group by core status, or show columns flat?	Grouped. Ten flat columns on a tablet is a scroll with no landmarks. Grouping under the six core statuses makes the locked structure visible, which is the thing the whole model rests on.
What does a Lead see?	The whole board, moving only his own cards. He needs to know what is coming. Already how the policies are written.
Drag on a phone?	No. Phone gets a tap-to-move action sheet. Dragging a card across ten columns one-handed on a ladder is a bad idea.
Optimistic or wait for the server?	Optimistic with rollback. A board that pauses on every drag feels broken on site Wi-Fi. But a refused move must visibly snap back.
Offline moves?	Not yet. Offline is Phase 3+ and belongs with the close-out queue, which is where losing data actually costs something.

5. Risks specific to this phase

RISK	HANDLING
The UI and the database disagree about what a role may do	The UI never decides — it reads the same matrix the policies implement, and a refusal from the server is always believed over the local guess. When they disagree the user sees an error, not a silent no-op.
Drag-and-drop is fiddly with gloves	Large cards, generous drop targets, and a tap-to-move path that does not require dragging at all. Test with the Leads before Phase 8, not after.

Ten columns is already too many	Grouping under core status buys room, but SPEC's own risk list says cap custom columns per project. That cap belongs in Phase 6a when column management is built.
A move that half-succeeds	Status and <code>board_column_id</code> change together or not at all — the database trigger already refuses a mismatch, so the app must send both in one update.

6. Checklist

- ☐ Board renders the seeded project, grouped by core status
- ☐ Cards show what a Super needs at a glance: title, area, system, who, hours, rework count
- ☐ Tapping a card opens its detail and its event history
- ☐ A move derives status from the target column, never the reverse
- ☐ A Lead cannot drag toward Accomplished — **the milestone**
- ☐ A refused move snaps back and explains itself
- ☐ Create a task; assign Foreman then Lead
- ☐ New database assertions for column-to-status derivation
- ☐ Screenshots cover the board as Super and as Lead
- ☐ CI green on all four jobs

Phase 2 is done when a Lead, signed in on an iPad, can see his whole job, move his own work forward, and finds the Accomplished column closed to him — with the database refusing him even if the screen ever forgets to.